

Skill Check 3: Building Better Code

Exam Agreement

This Skill Check is an individual assessment and you should not receive or offer help on it from any other human.

You may use any **physical** resource to complete the work. This includes:

- Any notes, code, slides, papers, or previous feedback from the instructor as long as they are on paper.
- Any books that you have with you.
- Any scholarly works such as papers that you have with you.

You may NOT use:

- Help from any other student or person. **This is an individual assessment.**
- Any digital resource that does not exist as a physical copy present in class.
- The use of generative artificial intelligence (e.g., ChatGPT).
- Asking for help on specific questions. (It is fine to ask for clarification!)
- Help from homework websites such as Course Hero or Chegg.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Chapman ID: _____

Signed: _____

Date: _____

Include this signed page as the first page of your submitted work.

Question 2: LOOPS

Create a for loop that prints out the first 10 letters of the alphabet.

[illegible]

Question 3: CONDITIONALS

Load in the iris dataset. Write a conditional that prints “large average sepal length.” if the average sepal length (Sepal.Length column) of plants in the dataset is more than 5, “medium average sepal length” if the average sepal length is > 3 but ≤ 5 , and “small average sepal length” otherwise.

[illegible]

Question 4: COMBINATIONS

You have a graph of displacement versus horsepower for cars in the `mtcars` data set. You're trying out different color and sizes of the points on the graph with the code below:

```
library(ggplot2)
data("mtcars")

ggplot(mtcars, aes(hp, disp)) +
  geom_point(color = "black", size = 2) +
  xlab("Horsepower") + ylab("Displacement") +
  theme_minimal()

ggplot(mtcars, aes(hp, disp)) +
  geom_point(color = "red", size = 4) +
  xlab("Horsepower") + ylab("Displacement") +
  theme_minimal()

ggplot(mtcars, aes(hp, disp)) +
  geom_point(color = "gold", size = 1) +
  xlab("Horsepower") + ylab("Displacement") +
  theme_minimal()
```

Write a function that will allow you to alter point size and color. Include a function call to your function that reproduces one of the example graphs above.

[illegible]